Natural Selection Experiment

Aim: To simulate the process of natural selection

Materials: 3 pieces of card, 1 white (snow), 1 brown (desert), 1 green (grassland)
1 set of 60 animals (30 brown and 30 white)
Blu-tac
Stopwatch

Method

1. Select a habitat.
2. Choose one person to be the hunter and one person to be the time keeper and recorder.
3. The hunter closes their eyes while the timekeeper spreads the animals out on the habitat.
4. Once the animals are laid out, the hunter will have 20 seconds to 'eat' as many animals as they can by picking them up with the Blu-tac and putting them in a small pile.
5. Count the number of animals of each type eaten and record it in the table below
6. Repeat the experiment twice more with the same habitat and calculate an average.
7. Change to a different habitat and repeat the process.

Results

Desert Habitat

	Trial 1	Trial 2	Trial 3	Average
No. of brown animals at start				
No. of brown animals eaten				
No. of white animals at start				
No. of white animals eaten				

Snow Habitat

	Trial 1	Trial 2	Trial 3	Average
No. of brown animals at start				
No. of brown animals eaten				
No. of white animals at start				
No. of white animals eaten				

Grassland Habitat

	Trial 1	Trial 2	Trial 3	Average
No. of brown animals at start				
No. of brown animals eaten				
No. of white animals at start				
No. of white animals eaten				

Conclusions and Questions

What do you notice about how many animals are eaten and the colour of the background habitat?

How can this be related to real animals and the adaptations they have?

If your animals were real animals in the wild, they would reproduce. Assuming each animal produces offspring with the same adaptations, in this case colour, what do you think would happen to the population over successive generations in each habitat?

In your own words, describe the process of natural selection.